

Analysis of Linear Programming

Part 3c: Unstructured LPs – Big-M and Two-Phase Methods

IE 311 – Operations Research I

- So far we have only discussed how to solve **maximization** problems with only $\leq$ constraints and a nonnegative right-hand-side vector b (an obvious initial BFS is then available).
- Now we learn how to incorporate $\geq$ and $=$ constraints with an arbitrary right-hand side. In this case an initial BFS may **not** be obvious.
- One of the following methods is used to obtain an initial BFS:
- the **Big-M Method**, or
- the **Two-Phase Method**.

Example 6

- Consider the following LP:

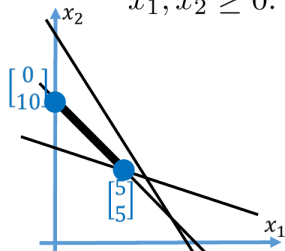
$$\max x_0 = 2x_1 + 3x_2$$

$$\text{s.t.} \quad 4x_1 + 2x_2 \leq 32$$

$$x_1 + 3x_2 \geq 20$$

$$x_1 + x_2 = 10$$

$$x_1, x_2 \geq 0.$$



- The standard form is as follows:

$$\max x_0 = 2x_1 + 3x_2$$

$$\text{s.t.} \quad 4x_1 + 2x_2 + x_3 = 32$$

$$x_1 + 3x_2 - x_4 = 20$$

$$x_1 + x_2 = 10$$

$$x_1, x_2, x_3, x_4 \geq 0.$$

- An initial basis is not obvious**
(the origin is not feasible)!

Example 6: A Modified Version

- Consider the modified constraints (notice the *artificial variables* x_5 and x_6):

$$4x_1 + 2x_2 + x_3 = 32$$

$$x_1 + 3x_2 - x_4 + x_5 = 20$$

$$x_1 + x_2 + x_6 = 10$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0.$$

- Now, an obvious basis is available (**WHY?**).
- However, the feasible region and, consequently, the LP are **different**.
- In order to make sure that the modified problem is equivalent to the original problem, we will **penalize** the artificial variables x_5 and x_6 .

- First, modify the constraints so that the right-hand side of each constraint is nonnegative.
- For each constraint:
- if it is a $\leq$ constraint, add a **slack** variable;
- if it is a $\geq$ constraint, subtract a **surplus** variable and add an **artificial** variable;
- if it is an $=$ constraint, add an **artificial** variable.
- To ensure the modified problem is equivalent to the original, we **penalize** all artificial variables.
- There are two different ways to do this: the **Big-M Method** and the **Two-Phase Method**.

Main idea: add the artificial variables to the objective with large negative coefficients ($-M$) and solve the resulting modified problem.

We then proceed with the following steps:

- Choose an appropriate initial basis using the slack and artificial variables.
- Zero-out the objective-row coefficients of the artificial variables.
- Solve the remaining problem with the simplex method.

If at least one artificial variable has a **positive** value at optimality, then the original problem is **infeasible**.

Example 6: Big-M Method

Let $M > 0$ be a large number.

Consider the modified problem:

$$\begin{aligned} \max x_0 &= 2x_1 + 3x_2 && - Mx_5 - Mx_6 \\ \text{s.t.} \quad &4x_1 + 2x_2 + x_3 && = 32 \\ &x_1 + 3x_2 &- x_4 + x_5 &= 20 \\ &x_1 + x_2 && + x_6 = 10 \\ &x_1, x_2, x_3, x_4, x_5, x_6 && \geq 0. \end{aligned}$$

We will now use the simplex method.

Example 6 (Big-M Method): Iteration 0

Row	BV							RHS
0	x_0	x_0	$-2x_1$	$-3x_2$		$+Mx_5$	$+Mx_6$	$= 0$
1	x_3		$4x_1$	$+2x_2$	$+x_3$			$= 32$
2	x_5		x_1	$+3x_2$		$-x_4$	$+x_5$	$= 20$
3	x_6		x_1	$+x_2$			$+x_6$	$= 10$

- We need to zero-out the objective coefficients of the artificial variables x_5 and x_6 first.

Row	BV							RHS
0	x_0	x_0	$-(2M + 2)x_1$	$-(4M + 3)x_2$		$+Mx_4$		$= -30M$
1	x_3		$4x_1$	$+2x_2$	$+x_3$			$= 32$
2	x_5		x_1	$+3x_2$		$-x_4$	$+x_5$	$= 20$
3	x_6		x_1	$+x_2$			$+x_6$	$= 10$

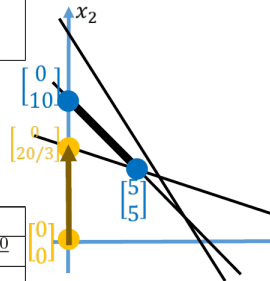
Example 6 (Big-M Method): Iteration 1

Row	BV						RHS
0	x_0	x_0	$-(2M+2)x_1$	$-(4M+3)x_2$	$+Mx_4$		$= -30M$
1	x_3		$4x_1$	$+2x_2$	$+x_3$		$= 32$
2	x_5		x_1	$+3x_2$	$-x_4$	$+x_5$	$= 20$
3	x_6		x_1	$+x_2$		$+x_6$	$= 10$

Entering Variable: x_2 (most negative coefficient = $-(4M+3)$)

Leaving Variable: x_5 (minimum ratio = $20/3$)

Row	BV						RHS
0	x_0	x_0	$-\frac{2M+3}{3}x_1$	$-\frac{M+3}{3}x_4$	$+\frac{4M+3}{3}x_5$		$= -\frac{10M-60}{3}$
1	x_3		$\frac{10}{3}x_1$	$+x_3$	$+\frac{2}{3}x_4$	$-\frac{2}{3}x_5$	$= \frac{56}{3}$
2	x_2		$\frac{1}{3}x_1$	$+x_2$	$-\frac{1}{3}x_4$	$+\frac{1}{3}x_5$	$= \frac{20}{3}$
3	x_6		$\frac{2}{3}x_1$		$+\frac{1}{3}x_4$	$-\frac{1}{3}x_5$	$+\frac{10}{3}x_6$



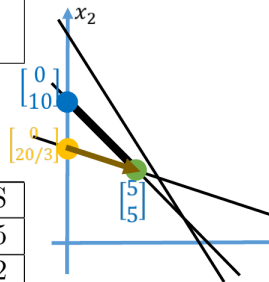
Example 6 (Big-M Method): Iteration 2

Row	BV						RHS
0	x_0	x_0	$-\frac{2M+3}{3}x_1$		$-\frac{M+3}{3}x_4$	$+\frac{4M+3}{3}x_5$	$= -\frac{10M-60}{3}$
1	x_3		$\frac{10}{3}x_1$	$+x_3$	$+\frac{2}{3}x_4$	$-\frac{2}{3}x_5$	$= \frac{56}{3}$
2	x_2		$\frac{1}{3}x_1$	$+x_2$	$-\frac{1}{3}x_4$	$+\frac{1}{3}x_5$	$= \frac{20}{3}$
3	x_6		$\frac{2}{3}x_1$		$+\frac{1}{3}x_4$	$-\frac{1}{3}x_5$	$+x_6 = \frac{10}{3}$

Entering Variable: x_1 (most negative coefficient = $-(2M + 3)/3$)

Leaving Variable: x_6 (minimum ratio = 5)

Row	BV						RHS
0	x_0	x_0		$-\frac{1}{2}x_4$	$+\frac{2M+1}{2}x_5$	$+\frac{2M+3}{2}x_6$	$= 25$
1	x_3		x_3	$-x_4$	$+x_5$	$-5x_6$	$= 2$
2	x_2		x_2	$-\frac{1}{2}x_4$	$+\frac{1}{2}x_5$	$-\frac{1}{2}x_6$	$= 5$
3	x_1		x_1	$+\frac{1}{2}x_4$	$-\frac{1}{2}x_5$	$+\frac{3}{2}x_6$	$= 5$



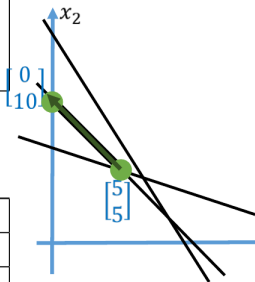
Example 6 (Big-M Method): Iteration 3

Row	BV					RHS
0	x_0	x_0	$-\frac{1}{2}x_4$	$+\frac{2M+1}{2}x_5$	$+\frac{2M+3}{2}x_6$	$= 25$
1	x_3	x_3	$-x_4$	$+x_5$	$-5x_6$	$= 2$
2	x_2	x_2	$-\frac{1}{2}x_4$	$+\frac{1}{2}x_5$	$-\frac{1}{2}x_6$	$= 5$
3	x_1	x_1	$+\frac{1}{2}x_4$	$-\frac{1}{2}x_5$	$+\frac{3}{2}x_6$	$= 5$

Entering Variable: x_4 (most negative coefficient = $-1/2$)

Leaving Variable: x_1 (minimum ratio = 10)

Row	BV					RHS
0	x_0	x_0	$+x_1$	$+Mx_5$	$+(M+3)x_6$	$= 30$
1	x_3	$2x_1$	$+x_3$		$-2x_6$	$= 12$
2	x_2	x_1	$+x_2$		$+x_6$	$= 10$
3	x_4	$2x_1$	$+x_4$	$-x_5$	$+3x_6$	$= 10$



Example 6 (Big-M Method): Iteration 4

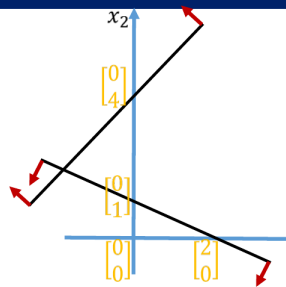
Row	BV						RHS
0	x_0	x_0	$+x_1$		$+Mx_5$	$+(M+3)x_6$	$= 30$
1	x_3		$2x_1$	$+x_3$		$-2x_6$	$= 12$
2	x_2		x_1	$+x_2$		$+x_6$	$= 10$
3	x_4		$2x_1$		$+x_4$	$-x_5$	$+3x_6$ $= 10$

- There is no entering variable. **STOP.**
- The value of all the artificial variables is 0.
 - The problem is feasible.
 - This is the optimal tableau (there is a unique optimal solution).

Example 7

- Use the simplex method to solve the **infeasible LP**:

$$\begin{aligned} \max \quad & x_0 = 2x_1 + 4x_2 \\ \text{s.t.} \quad & x_1 - x_2 \leq -4 \\ & x_1 + 2x_2 \leq 2 \\ & x_1, x_2 \geq 0. \end{aligned}$$



- We first obtain the modified LP: $\max x_0 = 2x_1 + 4x_2 - Mx_4$

$$\begin{aligned} \text{s.t.} \quad & -x_1 + x_2 - x_3 + x_4 = 4 \\ & x_1 + 2x_2 + x_5 = 2 \\ & x_1, x_2, x_3, x_4, x_5 \geq 0. \end{aligned}$$

Example 7 (Big-M Method): Iteration 0

Row	BV						RHS
0	x_0	x_0	$-2x_1$	$-4x_2$		$+Mx_4$	$= 0$
1	x_4		$-x_1$	$+x_2$	$-x_3$	$+x_4$	$= 4$
2	x_5		x_1	$+2x_2$		$+x_5$	$= 2$

We first zero-out the objective coefficient of x_4 .

Row	BV						RHS
0	x_0	x_0	$+(M-2)x_1$	$-(M+4)x_2$	$+Mx_3$		$= -4M$
1	x_4		$-x_1$	$+x_2$	$-x_3$	$+x_4$	$= 4$
2	x_5		x_1	$+2x_2$		$+x_5$	$= 2$

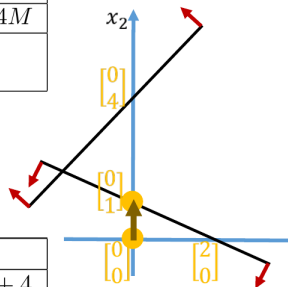
Example 7 (Big-M Method): Iteration 1

Row	BV						RHS
0	x_0	x_0	$+(M-2)x_1$	$-(M+4)x_2$	$+Mx_3$		$= -4M$
1	x_4		$-x_1$	$+x_2$	$-x_3$	$+x_4$	$= 4$
2	x_5		x_1	$+2x_2$		$+x_5$	$= 2$

Entering Variable: x_2 (most negative coefficient $= -(M+4)$)

Leaving Variable: x_5 (minimum ratio $= 1$)

Row	BV						RHS
0	x_0	x_0	$+\frac{3M}{2}x_1$		$+Mx_3$	$+\frac{M+4}{2}x_5$	$= -3M + 4$
1	x_4		$-\frac{3}{2}x_1$		$-x_3$	$+\frac{1}{2}x_5$	$= 3$
2	x_2		$\frac{1}{2}x_1$	$+x_2$		$+\frac{1}{2}x_5$	$= 1$



Example 7 (Big-M Method): Iteration 2

Row	BV					RHS
0	x_0	x_0	$+\frac{3M}{2}x_1$	$+Mx_3$	$+\frac{M+4}{2}x_5$	$= -3M + 4$
1	x_4		$-\frac{3}{2}x_1$	$-x_3$	$+x_4$	$= 3$
2	x_2		$+\frac{1}{2}x_1$	$+x_2$	$+\frac{1}{2}x_5$	$= 1$

- There is no entering variable. **STOP**.
- The value of the artificial variable x_4 is positive.
 - The problem is INFEASIBLE.

Main steps:

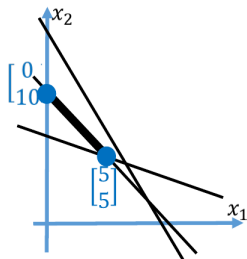
- Solve an auxiliary problem (**Phase I**) whose objective is to *minimize the sum of the artificial variables*, using the simplex method.
- If the optimal value of the auxiliary problem is **positive**, STOP: the original problem is **infeasible**.
- If some artificial variables are still basic at the end of Phase I (they must be at value 0 – why?), pivot them out of the basis; once all artificial variables are nonbasic, drop them from the tableau.
- Finally, bring back the original objective and solve the original problem (**Phase II**), starting from the basis obtained at the end of Phase I.

- Consider the following LP:

$$\begin{aligned} \max x_0 &= 2x_1 + 3x_2 \\ \text{s.t.} \quad &4x_1 + 2x_2 \leq 32 \\ &x_1 + 3x_2 \geq 20 \\ &x_1 + x_2 = 10 \end{aligned}$$

- We first solve the auxiliary (Phase I) LP:

$$\begin{aligned} -\max x'_0 &= -x_5 - x_6 \\ \text{s.t.} \quad &4x_1 + 2x_2 + x_3 = 32 \\ &x_1 + 3x_2 - x_4 + x_5 = 20 \\ &x_1 + x_2 + x_6 = 10 \\ &x_1, x_2, x_3, x_4, x_5, x_6 \geq 0. \end{aligned}$$



Example 6 (Phase I): Iteration 0

Row	BV					RHS
0	x'_0	x'_0		$+x_5$	$+x_6$	$= 0$
1	x_3	$4x_1$	$+2x_2$	$+x_3$		$= 32$
2	x_5	x_1	$+3x_2$	$-x_4$	$+x_5$	$= 20$
3	x_6	x_1	$+x_2$		$+x_6$	$= 10$

- We need to zero-out the objective coefficients of the artificial variables x_5 and x_6 first.

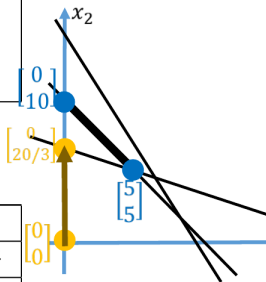
Row	BV					RHS
0	x'_0	x'_0	$-2x_1$	$-4x_2$	$+x_4$	$= -30$
1	x_3	$4x_1$	$+2x_2$	$+x_3$		$= 32$
2	x_5	x_1	$+3x_2$	$-x_4$	$+x_5$	$= 20$
3	x_6	x_1	$+x_2$		$+x_6$	$= 10$

Example 6 (Phase I): Iteration 1

Row	BV						RHS
0	x'_0	x'_0	$-2x_1$	$-4x_2$	$+x_4$		$= -30$
1	x_3		$4x_1$	$+2x_2$	$+x_3$		$= 32$
2	x_5		x_1	$+3x_2$	$-x_4$	$+x_5$	$= 20$
3	x_6		x_1	$+x_2$		$+x_6$	$= 10$

- **Entering Variable:** x_2 (most negative coefficient = -4)
- **Leaving Variable:** x_5 (minimum ratio = $20/3$)

Row	BV						RHS
0	x'_0	x'_0	$-\frac{2}{3}x_1$		$-\frac{1}{3}x_4$	$+\frac{4}{3}x_5$	$= -\frac{10}{3}$
1	x_3		$\frac{10}{3}x_1$	$+x_3$	$+\frac{2}{3}x_4$	$-\frac{2}{3}x_5$	$= \frac{56}{3}$
2	x_2		$\frac{1}{3}x_1$	$+x_2$	$-\frac{1}{3}x_4$	$+\frac{1}{3}x_5$	$= \frac{20}{3}$
3	x_6		$\frac{2}{3}x_1$		$+\frac{1}{3}x_4$	$-\frac{1}{3}x_5$	$+x_6$ $= \frac{10}{3}$

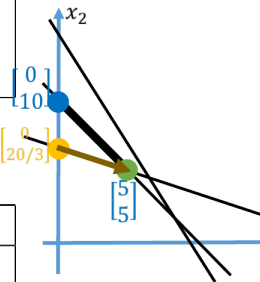


Example 6 (Phase I): Iteration 2

Row	BV						RHS
0	x'_0	x'_0	$-\frac{2}{3}x_1$		$-\frac{1}{3}x_4$	$+\frac{4}{3}x_5$	$= -\frac{10}{3}$
1	x_3		$\frac{10}{3}x_1$	$+x_3$	$+\frac{2}{3}x_4$	$-\frac{2}{3}x_5$	$= \frac{56}{3}$
2	x_2		$\frac{1}{3}x_1$	$+x_2$	$-\frac{1}{3}x_4$	$+\frac{1}{3}x_5$	$= \frac{20}{3}$
3	x_6		$\frac{2}{3}x_1$		$+\frac{1}{3}x_4$	$-\frac{1}{3}x_5$	$+x_6 = \frac{10}{3}$

- **Entering Variable:** x_1 (most negative coefficient = $-2/3$)
- **Leaving Variable:** x_6 (minimum ratio = 5)

Row	BV						RHS
0	x'_0	x'_0			$+x_5$	$+x_6$	$= 0$
1	x_3		x_3	$-x_4$	$+x_5$	$-5x_6$	$= 2$
2	x_2		x_2	$-\frac{1}{2}x_4$	$+\frac{1}{2}x_5$	$-\frac{1}{2}x_6$	$= 5$
3	x_1		x_1	$+\frac{1}{2}x_4$	$-\frac{1}{2}x_5$	$+\frac{3}{2}x_6$	$= 5$



Example 6 (Phase I): Iteration 3

Row	BV					RHS	
0	x'_0	x'_0		$+x_5$	$+x_6$	$= 0$	
1	x_3		x_3	$-x_4$	$+x_5$	$-5x_6$	$= 2$
2	x_2		x_2	$-\frac{1}{2}x_4$	$+\frac{1}{2}x_5$	$-\frac{1}{2}x_6$	$= 5$
3	x_1		x_1	$+\frac{1}{2}x_4$	$-\frac{1}{2}x_5$	$+\frac{3}{2}x_6$	$= 5$

- There is no entering variable. **STOP.**
- The value of all the artificial variables is 0.
 - The problem is feasible.
 - All the artificial variables are nonbasic.
 - The optimal basis of Phase I LP can be used as the initial basis for the original problem (Phase II LP).

Example 6 (Phase II): Iteration 0

Row	BV				RHS	
0	x_0	x_0	$-2x_1$	$-3x_2$	$= 0$	
1	x_3			x_3	$-x_4$	$= 2$
2	x_2			x_2	$-\frac{1}{2}x_4$	$= 5$
3	x_1		x_1		$+\frac{1}{2}x_4$	$= 5$

- We need to zero-out the objective coefficients of the variables x_1 , x_2 and x_3 first.

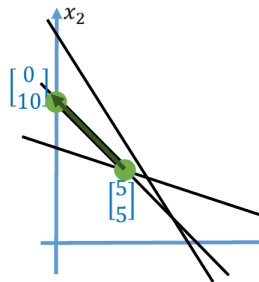
Row	BV				RHS
0	x_0	x_0		$-\frac{1}{2}x_4$	$= 25$
1	x_3		x_3	$-x_4$	$= 2$
2	x_2		x_2	$-\frac{1}{2}x_4$	$= 5$
3	x_1		x_1	$+\frac{1}{2}x_4$	$= 5$

Example 6 (Phase II): Iteration 1

Row	BV			RHS
0	x_0	x_0	$-\frac{1}{2}x_4$	$= 25$
1	x_3	x_3	$-x_4$	$= 2$
2	x_2	x_2	$-\frac{1}{2}x_4$	$= 5$
3	x_1	x_1	$+\frac{1}{2}x_4$	$= 5$

- **Entering Variable:** x_4 (most negative coefficient $= -1/2$)
- **Leaving Variable:** x_1 (minimum ratio $= 10$)

Row	BV			RHS
0	x_0	x_0	$+x_1$	$= 30$
1	x_3	$2x_1$	$+x_3$	$= 12$
2	x_2	x_1	$+x_2$	$= 10$
3	x_4	$2x_1$	$+x_4$	$= 10$



Example 6 (Phase II): Iteration 2

Row	BV			RHS
0	x_0	x_0	$+x_1$	$= 30$
1	x_3	$2x_1$	$+x_3$	$= 12$
2	x_2	x_1	$+x_2$	$= 10$
3	x_4	$2x_1$	$+x_4$	$= 10$

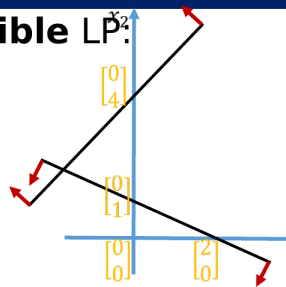
- There is no entering variable. **STOP.**
- This is the optimal tableau (there is a unique optimal solution).

- Use the simplex method to solve the **infeasible** LP:

$$\begin{aligned} \max x_0 &= 2x_1 + 4x_2 \\ \text{s.t.} \quad &x_1 - x_2 \leq -4 \\ &x_1 + 2x_2 \leq 2 \\ &x_1, x_2 \geq 0. \end{aligned}$$

- We first solve the auxiliary (Phase I) LP:

$$\begin{aligned} -\max x'_0 &= -x_4 \\ \text{s.t.} \quad &-x_1 + x_2 - x_3 + x_4 = 4 \\ &x_1 + 2x_2 + x_5 = 2 \\ &x_1, x_2, x_3, x_4, x_5 \geq 0. \end{aligned}$$



Example 7 (Phase I): Iteration 0

Row	BV					RHS	
0	x'_0	x'_0			$+x_4$	$= 0$	
1	x_4		$-x_1$	$+x_2$	$-x_3$	$+x_4$	$= 4$
2	x_5		x_1	$+2x_2$		$+x_5$	$= 2$

- We first zero-out the objective coefficient of x_4 .

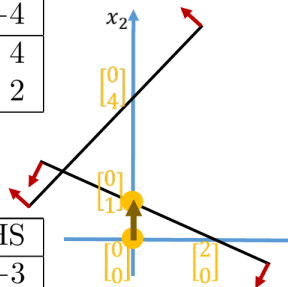
Row	BV						RHS
0	x'_0	x'_0	$+x_1$	$-x_2$	$+x_3$		$= -4$
1	x_4		$-x_1$	$+x_2$	$-x_3$	$+x_4$	$= 4$
2	x_5		x_1	$+2x_2$		$+x_5$	$= 2$

Example 7 (Phase I): Iteration 1

Row	BV						RHS
0	x'_0	x'_0	$+x_1$	$-x_2$	$+x_3$		$= -4$
1	x_4		$-x_1$	$+x_2$	$-x_3$	$+x_4$	$= 4$
2	x_5		x_1	$+2x_2$		$+x_5$	$= 2$

- **Entering Variable:** x_2 (most negative coefficient = -1)
- **Leaving Variable:** x_5 (minimum ratio = 1)

Row	BV						RHS
0	x'_0	x'_0	$+\frac{3}{2}x_1$		$+x_3$	$+\frac{1}{2}x_5$	$= -3$
1	x_4		$-\frac{3}{2}x_1$		$-x_3$	$+\frac{1}{2}x_5$	$= 3$
2	x_2		$\frac{1}{2}x_1$	$+x_2$		$+\frac{1}{2}x_5$	$= 1$



Example 7 (Phase I): Iteration 2

Row	BV						RHS
0	x'_0	x'_0	$+\frac{3}{2}x_1$	$+x_3$	$+\frac{1}{2}x_5$		$= -3$
1	x_4		$-\frac{3}{2}x_1$	$-x_3$	$+x_4$	$-\frac{1}{2}x_5$	$= 3$
2	x_2		$\frac{1}{2}x_1$	$+x_2$		$+\frac{1}{2}x_5$	$= 1$

- There is no entering variable. **STOP.**
- The optimal value of Phase I LP is positive.
 - The problem is INFEASIBLE.
 - We do NOT proceed with Phase II.

- Use the simplex method to solve the following LP:

$$\max x_0 = 3x_1 + 2x_2 + 3x_3$$

$$\text{s.t.} \quad 2x_1 + x_2 + x_3 \leq 2$$

$$3x_1 + 4x_2 + 2x_3 \geq 8$$

$$x_1, x_2, x_3 \geq 0.$$

- We first solve the auxiliary (Phase I) LP:

$$-\max x'_0 = \quad \quad \quad -x_6$$

$$\text{s.t.} \quad 2x_1 + x_2 + x_3 + x_4 \quad \quad \quad = 2$$

$$3x_1 + 4x_2 + 2x_3 \quad \quad - x_5 + x_6 = 8$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0.$$

Example 8 (Phase I): Iteration 0

Row	BV							RHS
0	x'_0	x'_0					$+x_6$	$= 0$
1	x_4		$2x_1$	$+x_2$	$+x_3$	$+x_4$		$= 2$
2	x_6		$3x_1$	$+4x_2$	$+2x_3$		$-x_5 + x_6$	$= 8$

- We first zero-out the objective coefficient of x_6 .

Row	BV							RHS
0	x'_0	x'_0	$-3x_1$	$-4x_2$	$-2x_3$		$+x_5$	$= -8$
1	x_4		$2x_1$	$+x_2$	$+x_3$	$+x_4$		$= 2$
2	x_6		$3x_1$	$+4x_2$	$+2x_3$		$-x_5 + x_6$	$= 8$

Example 8 (Phase I): Iteration 1

Row	BV							RHS
0	x'_0	x'_0	$-3x_1$	$-4x_2$	$-2x_3$		$+x_5$	$= -8$
1	x_4		$2x_1$	$+x_2$	$+x_3$	$+x_4$		$= 2$
2	x_6		$3x_1$	$+4x_2$	$+2x_3$		$-x_5$ $+x_6$	$= 8$

- **Entering Variable:** x_2 (most negative coefficient = -4)
- **Leaving Variable:** x_4 (minimum ratio = 2- **there is a tie**)

Row	BV							RHS
0	x'_0	x'_0	$+5x_1$		$+2x_3$	$+4x_4$	$+x_5$	$= 0$
1	x_2		$2x_1$	$+x_2$	$+x_3$	$+x_4$		$= 2$
2	x_6		$-5x_1$		$-2x_3$	$-4x_4$	$-x_5$ $+x_6$	$= 0$

Example 8 (Phase I): Iteration 2

Row	BV							RHS
0	x'_0	x'_0	$+5x_1$		$+2x_3$	$+4x_4$	$+x_5$	$= 0$
1	x_2		$2x_1$	$+x_2$	$+x_3$	$+x_4$		$= 2$
2	x_6		$-5x_1$		$-2x_3$	$-4x_4$	$-x_5$	$+x_6$ $= 0$

- There is no entering variable. **STOP**.
- The value of the artificial variable is 0.
 - The problem is feasible.
 - Note that the artificial variables x_6 is basic.

Example 8 (Phase II): Iteration 0

Row	BV							RHS
0	x_0	x_0	$-3x_1$	$-2x_2$	$-3x_3$			$= 0$
1	x_2		$2x_1$	$+x_2$	$+x_3$	$+x_4$		$= 2$
2	x_6		$-5x_1$		$-2x_3$	$-4x_4$	$-x_5$	$+x_6$ $= 0$

- We need to zero-out the objective coefficients of the variables x_2 and x_6 first.
- Recall that the artificial variables x_6 is basic.
- We need to make x_6 leave the basis. Let us *force* x_1 enter the basis (we could have chosen any variable whose coefficient in row-2 is nonzero) and x_6 leave the basis.

Example 8 (Phase II): Iteration 1

Row	BV					RHS
0	x_0	x_0	$-\frac{7}{5}x_3$	$+\frac{6}{5}x_4$	$-\frac{1}{5}x_5$	$= 4$
1	x_2	x_2	$+\frac{1}{5}x_3$	$-\frac{3}{5}x_4$	$-\frac{2}{5}x_5$	$= 2$
2	x_1	x_1	$+\frac{2}{5}x_3$	$+\frac{4}{5}x_4$	$+\frac{1}{5}x_5$	$= 0$

- **Entering Variable:** x_3 (most negative coefficient = $-7/5$)
- **Leaving Variable:** x_1 (minimum ratio = 0- **degenerate pivot**)

Row	BV					RHS
0	x_0	x_0	$+\frac{7}{2}x_1$	$+4x_4$	$+\frac{1}{2}x_5$	$= 4$
1	x_2	$-\frac{1}{2}x_1$	$+x_2$	$-x_4$	$-\frac{1}{2}x_5$	$= 2$
2	x_3	$\frac{5}{2}x_1$	$+x_3$	$+2x_4$	$+\frac{1}{2}x_5$	$= 0$

Example 8 (Phase II): Iteration 1

Row	BV						RHS
0	x_0	x_0	$+\frac{7}{2}x_1$		$+4x_4$	$+\frac{1}{2}x_5$	$= 4$
1	x_2		$-\frac{1}{2}x_1$	$+x_2$	$-x_4$	$-\frac{1}{2}x_5$	$= 2$
2	x_3		$+\frac{5}{2}x_1$	$+x_3$	$+2x_4$	$+\frac{1}{2}x_5$	$= 0$

- There is no entering variable. **STOP**.
- This is the optimal tableau (there is a unique optimal solution).

- Use the Python file `SimplexTableauWithBV.py` (uploaded to SUCourse) to solve Examples 6, 7 and 8 using the **Big-M Method**.
- Try to implement the **Two-Phase Method** using the file `SimplexTableau_Template.py` as a subroutine.